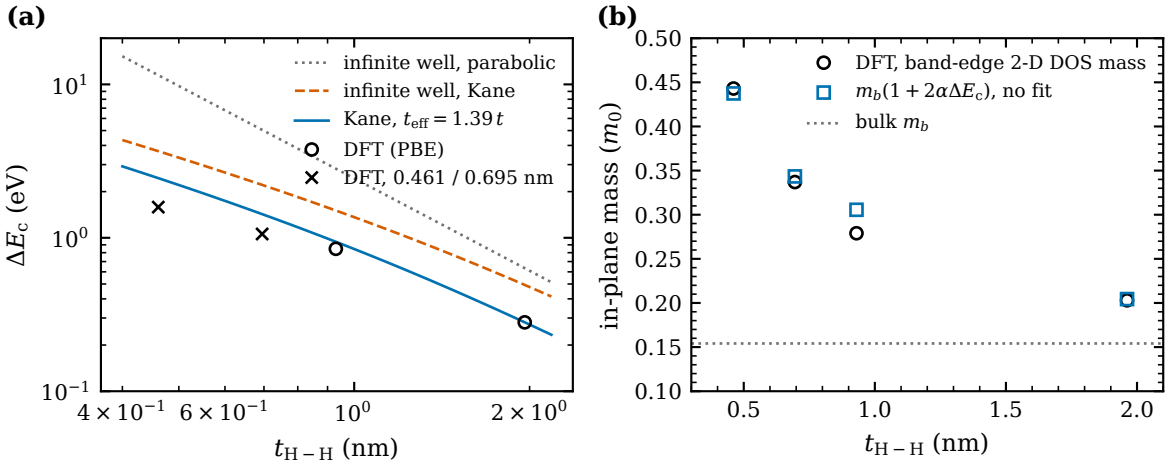
(a)
infinite well, parabolic
infinite well, Kane
Kane, $t_{\mathrm{eff}} = 1.39\,t$
DFT (PBE)
DFT, 0.461 / 0.695 nm
ΔE_c (eV)
10^1
10^0
10^{-1}
4×10^{-1}
6×10^{-1}
10^0
2×10^0
$t_{\mathrm{H-H}}$ (nm)
(b)
DFT, band-edge 2-D DOS mass
$m_b(1 + 2\alpha\Delta E_c)$, no fit
bulk m_b
in-plane mass (m_0)
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.5
1.0
1.5
2.0
$t_{\mathrm{H-H}}$ (nm)